to a unique rref. By Theorem 2, the matrix A has no free variables. By defn, all the n columns of the matrix are pivot columns. By defn. of a pivot, n rows of the matrix have pivots. $\boxed{\therefore m \geq n}$

Each statement in Exercises 33–38 is either true (in all cases) or false (for atleast one example). If false, construct a specific example to show that the statement is not always true. Such an example is called a $\boxed{\text{counterexample}}$ to the statement. If a statement is true, give a justification. (One specific example cannot explain why a statement is always true. You will have to do more work here than in Exercises 21 and 22.)

33. If $v_1, \ldots, v_4$ are in $\mathbb{R}^4$ and $v_3 = 2v_1 + v_2$, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.

$\boxed{\text{Solution:}}$ $v_3 = 2v_1 + v_2 \Rightarrow v_3 = (2v_1 + v_2) + 0$ (Algebraic property iii)

$\Rightarrow v_3 = 2v_1 + 1 \cdot v_2 + 0 \cdot v_4$ (Algebraic property viii)

By defn, v_3 is a linear combination of v_1, v_2 and v_4. From Theorem 7, $\{v_1, v_2, v_3, v_4\}$ is linearly dependent. $\boxed{\therefore \text{The statement is True}}$.

34. If $v_1, \ldots, v_4$ are in $\mathbb{R}^4$ and $v_3 = 0$, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.

$\boxed{\text{Solution:}}$ $\because v_3 = 0$, By Theorem 9, $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.

$\boxed{\therefore \text{The statement is True}}$.

35. If v_1 and v_2 are in $\mathbb{R}^4$ and v_2 is not a scalar multiple of v_1, then $\{v_1, v_2\}$ is linearly independent.

$\boxed{\text{Solution:}}$ From the topmost grey box of Pg-59, we know the given statement $\therefore$ is True.

~~35~~. The answer should be actually False, because one of the vectors can be a zero vector. Then By Theorem 9, $\{v_1, v_2\}$ must be linearly dependent.

36. If $v_1, \ldots, v_4$ are in $\mathbb{R}^4$ and v_3 is not a linear combination of v_1, v_2, v_4, then $\{v_1, v_2, v_3, v_4\}$ is linearly independent.

$\boxed{\text{Solution:}}$ Let $v_3 \neq 0$ and $v_1 = v_2 = v_4 = 0$. $\therefore \forall c_1, c_2, c_3 \in \mathbb{R}$, $c_1v_1 + c_2v_2 + c_3v_4 = 0 \neq v_3$

$\therefore v_3$ is not a linear combination of $\{v_1, v_2, v_4\}$. By Theorem 9, $\{v_1, v_2, v_3, v_4\}$ is linearly dependent. $\boxed{\therefore \text{The given statement is False}}$.